

FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

Multiparty Set Reconciliation

Pedro Vidal Marcelino

WORKING VERSION



Mestrado em Engenharia Informática e Computação

Supervisor: Prof. Carlos Baquero-Moreno

March 17, 2026

Contents

1	Approach	1
1.1	Overview	1
1.2	System Model	1
1.3	Evaluated protocols	2
1.3.1	Hybrid Rateless Bloom Filter with Rateless IBLT	2
1.3.2	RIBLT	2
1.3.3	Static Bloom Filter with IBLT	2
1.3.4	Full state transfer	2
1.4	Multi party coordination strategies	3
1.4.1	Pairwise Reconciliation	3
1.4.2	Topology Aware Reconciliation	3
1.5	Parameters	3
1.5.1	System parameters	3
1.5.2	Workload parameters	4
1.5.3	Execution Parameters	5
1.6	Metrics	5
1.6.1	Communication cost	5
1.6.2	Communication efficiency	6
1.6.3	Computation cost	6
1.6.4	Convergence behavior	6
1.6.5	Load distribution	6
1.6.6	Correctness	7
1.7	Work Plan	7
	References	9

Chapter 1

Approach

1.1 Overview

The objective of this thesis is to assess the rateless bloom filter as a reconciliation technique in multi replica environments. RBFs, have shown optimal communication cost in pairwise reconciliation [1], however these results are yet to be explored in scenarios with more than two replicas.

To assess the protocol's efficiency we are implementing a controlled experimental framework to simulate set reconciliation scenarios under a variety of workloads, topologies and set differences. It will be implemented in Rust due to its low level performance and memory safety features, which make it suitable to implement BF's and IBLT's efficiently with little computation overhead.

The evaluation will be conducted using a simulator that will execute every protocol participant on a single machine, as well as mimic real network communication by serializing messages and accounting for their bite size.

1.2 System Model

For each measured run, N replicas, each holding a set S_i will be simulated, the goal is for each replica to reconcile its state with every other replica, or in other words, to converge their state into the union of all the other sets, rendering the symmetric difference between each of them null. A run is only considered successful if and only if the following condition is met.

$$\forall i \in \{1, \dots, N\}, \quad S_i(t) \xrightarrow{t \rightarrow \infty} \bigcup_{j=1}^N S_j(0)$$

Each simulation will work under the assumption that there are no nodes with byzantine behavior as fault tolerance is outside of the scope of this thesis. Furthermore we will assume each replica won't change its state once the reconciliation process begins. This assumption keeps the system realistic as it is achievable in production scenarios by creating a snapshot of the current state once the process begins. Across every run, the network topology will guarantee every replica is accessible.

Both the communication and computation cost will be measured in a systematic way. Communication cost will be measured as the total number of bytes transmitted, this encompasses both protocol metadata and serialized elements. The computation cost is measured as the time spent in encoding and decoding operations by each replica.

1.3 Evaluated protocols

This section describes the pairwise reconciliation protocols that will be used in a two replica scenario as a way of calibrating the results of the following multi replica experiments.

1.3.1 Hybrid Rateless Bloom Filter with Rateless IBLT

This protocol is included as it is the main protocol to be evaluated in this thesis. It represents the state of the art for pairwise conciliation algorithms for sets with elements of varied size, without neither prior context nor any kind of parameter tuning. It takes advantage of a probabilistic first phase that uses a RBF to find likely common elements. It then utilizes a RIBLT to deterministically finish the reconciliation process, once the cost of reconciling confirmed negatives is outweighed by the cost of reconciling those same elements.

1.3.2 RIBLT

The purely RIBLT protocol will also be implemented and monitored as a baseline to assess the contribution of the RBF prefiltering in the first approach. This protocol will only use a RIBLT incrementally transmitting coded symbols from one replica to the other until the latter's peeling process succeeds, ensuring exact reconciliation.

1.3.3 Static Bloom Filter with IBLT

Despite not being up to date with the state of the art, this protocol will still be tested in a pairwise scenario in order to compare the cost (communication and computation) in protocols that require prior knowledge of d and those that don't (1.3.1). The protocol must first estimate d in order to accurately parameterize both the BF and the IBLT. Similarly to the hybrid rateless approach, the BF is used to find common elements beforehand, improving the IBLT's efficiency in retrieving the rest of the symmetric difference elements.

1.3.4 Full state transfer

This naive approach, despite simple to implement and inefficient when compared to others, will be simulated as an upper bound on communication cost and therefore provide a reference point against which the rest of the protocols can be evaluated.

1.4 Multi party coordination strategies

This section focus on assessing the impact on communication and computational efficiency, coordination has when performing set reconciliation in different multi replica scenarios with various common topologies and network latencies. Two different strategies will be applied in this thesis. Each model is to be assessed with two protocols: Hybrid RBF and RIBLT and pure RIBLT.

1.4.1 Pairwise Reconciliation

This is the naive approach that will be tested as the baseline model. Each replica will independently reconcile with every other replica. As mentioned in the state of the art it is expected to be communication heavy with a lot of redundant transmissions. In this model each replica will reconcile independently with every other replica, as mentioned before in the state of the art. Its main job is to define the upper overhead bound.

1.4.2 Topology Aware Reconciliation

This approach intends on taking advantage of the linearity provided by the improvements on IBLTs to support set reconciliation with multiple replicas shown in the state of the art section, without requiring multiple pairwise operations in the same network node. Replicas transmit their coded symbols across the network and intermediate nodes are responsible for calculating the total set difference in one go. Final reconciliation is performed only at designated nodes once sufficient information has been collected.

This strategy is included to evaluate the benefits of coordination in set reconciliation and how its scalability can be improved. To ensure correctness in the presence of multiple contributors, coded symbol aggregation will be performed over an appropriate finite field tailored to support linear subtraction, rather than using bitwise XOR that would only be capable of identifying element presence parity across replicas making it insufficient for this type of scenarios. Both mentioned protocols will need to be changed to reflect this change on BF and IBLTs.

1.5 Parameters

The experimental setup is going to be used to make many simulations each with different parameters. We have separated them based on what part of the protocol it is to be assessed as shown in 1.1. Table 1.1 shows exactly how this parameters will be put together in each simulation that is run. These parameters configuration will work as a guideline, and will later be adapted as simulations' data is acquired to better fit the protocols' requirements.

1.5.1 System parameters

These parameters are used to define the topology and size of the global environment the reconciliation takes place. The number of replicas $N \in \{2, 4, 8, 16, 32, 64\}$ is a wide range to evaluate how

Table 1.1: Experimental parameters and evaluated ranges. Baseline values are shown in bold.

Category	Parameter	Values / Description
System	Number of replicas (N)	2, 4, 8, 16 , 32, 64
	Topology	Star, Tree, Chord
Workload	Replica set size ($ S_i $)	10k, 100k , 500k elements
	Payload size	32 B, 256 B , 1 kB, 4 kB per element
	Digest size	64-bit , 128-bit
	Divergence level (δ)	0.1%, 1%, 5% , 10%, 20%
	Divergence pattern	Uniform random , clustered
	Cluster	Inter-cluster: 10–20%
Protocols	IBLT cells per increment	512, 1024 , 2048
Execution	Round cap	50
	Number of simulations	10

each protocol’s metrics grow as participation increases. As for the topology our aim was to accurately mimic most real world scenarios that may influence the execution of the protocols: star, tree and chord. Star represents scenarios with central bottlenecks, tree represents decentralized coordination environments with logarithmic routing depth. Finally chord, represent more decentralized topologies where many rounds of communication are required to reach every node. The chord topology was chosen over a gossip network due to the existence of node churn and high volatility in the latter since this is outside the scope of this thesis.

1.5.2 Workload parameters

Workload parameters represent the characteristics of the sets of each replica state. For instance, its size, the divergence level and pattern (clusters or not) as well as the digest and payload size used. Replica set sizes are varied

Workload parameters control the characteristics of replica state and the degree of disagreement that reconciliation must resolve. Replica set sizes are varied ($|S_i| \in \{10k, 100k, 500k\}$) which gives a wide range to measure the protocol’s efficiency both when metadata or payload is larger. Because elements may be variable-sized while reconciliation operates on fixed-size identifiers, payload sizes are varied (32 B to 4 kB) to quantify the extent to which “total bytes” is driven by the completion phase that transmits missing elements. Finally, we vary set divergence both in intensity and structure. The level of divergence $\delta \in \{0.1\%, 1\%, 5\%, 10\%, 20\%\}$ directly affects the workload to achieve reconciliation, however in real case scenarios, closer replicas have more similar sets, reason why testing uniformly randomly spread states need to be tested as well as clustered nodes, where intra-cluster divergence is smaller than inter-cluster.

1.5.3 Execution Parameters

This section is dedicated to setting how each simulation will be run and how reproducibility of the results is ensured. Each parameter configuration will be evaluated with multiple seeds (10), to average out the randomness introduced with hash distributions. A round cap will also be introduced in order to prevent niche situations where a protocol fails the reconciliation process or is being slow, from skewing the results. This occasions, will be reported as non convergent, and data will still be retrieved as an informative outcome when comparing strategies.

Table 1.2: Baseline configuration used unless stated otherwise.

Parameter	Baseline value
Replicas N	16
Topologies	star, tree, chord
Set size $ S $	100k elements
Payload / digest	256B payload, 64-bit digest
Divergence δ	5% (uniform random)
Protocol defaults	IBLT cells/increment = 1024
Repetitions	10 independent runs (seeds)
Round cap	50

Table 1.3: Experimental matrix. All parameters follow the baseline (Table 1.2) unless stated otherwise.

Block	Description	Values
1	Scaling in number of replicas N	$N \in \{4, 8, 16, 32, 64\}$
2	Divergence sensitivity	$\delta \in \{0.1\%, 1\%, 5\%, 10\%, 20\%\}$
3	Set-size scaling	$ S \in \{10k, 100k, 500k\}$
4	Divergence structure	inter-cluster $\delta \in [10\%, 20\%]$
5	Rateless IBLT sensitivity	cells/increment $\in \{512, 1024, 2048\}$

1.6 Metrics

Each simulation will be evaluated based on the following metrics.

1.6.1 Communication cost

The total number of bytes transmitted during the reconciliation process. It includes all the coordination messages, Bloom filter slices and IBLT coded symbols which provides a topology

independent metric that accurately translates to the protocol efficiency.

1.6.2 Communication efficiency

The total communication cost in itself is not useful to assess the efficiency of the protocol, therefore for every simulation it will be calculated the communication efficiency as the ration between the communication cost and the the theoretical minimum required to reconcile every set. To contextualize absolute communication cost, results are additionally expressed as a ratio between total transmitted data and the theoretical minimum required to reconcile only the set differences. This normalized metric allows comparison across different This ratio provides a normalized measure of efficiency, allowing comparison across different workloads.

1.6.3 Computation cost

Besides communication cost, computation is another important Computation cost is measured as the wall-clock time spent in protocol encoding and decoding operations at each replica. Encoding time captures the cost of generating reconciliation data structures, while decoding time captures the cost of interpreting received data and reconstructing missing elements. These measurements allow evaluation of how reconciliation overhead scales with the number of replicas and the chosen coordination strategy.

1.6.4 Convergence behavior

In multi party scenarios, convergence behavior can be defined as the number of reconciliation rounds required until correctness is achieved. For pairwise strategies, a round consists on a complete execution of a reconciliation algorithm across all selected edges. When it comes to protocols that do not require nodes to reconcile theirs sets two at a time, a round become becomes one full aggregation and decoding cycle, after which each replica has their local state updated accordingly.

In both cases, reconciliation is an iterative process, which facilitates the recording of the metric. At the start of which simulation, each replica i will be initialized with a set $S_i(0)$. The simulator then proceeds with reconciliation rounds until convergence is achieved at which point the measure is returned.

1.6.5 Load distribution

One crucial fact in this type of protocols is the strain and computation cost imposed on specific nodes such as intermediate ones or highly connected peers. This calls for measuring of the communication cost and computation cost on a per node basis, allowing to assess how different coordination strategies inflict disproportionate work on certain nodes.

1.6.6 Correctness

This metric is evaluated by verifying all replicas converge to identical sets containing exactly the union of their initial elements, and provides a boolean value to whether the reconciliation was successful or not.

1.7 Work Plan

This section presents the tasks planned for the completion of the proposed solution, as well as their distribution represented in a gantt chart in figure 1.1.

The tasks are divided in four phases. Firstly, the experimental environment setup is to be built within three weeks, this phase encompasses building the simulator and being able to run a single run with measurements. Following the setup phase there is a two week period equally divided between the implementation of both the pairwise and multi party protocols. After the first two phases, three weeks were allocated to calibrating the protocols (based on the pairwise baseline), computing each metric, setting up the workload scenarios and finally running every experiment and storing the results. The work plan ends with the results analysis and review period, to which two weeks were allocated.

1. Multi replica simulator
2. Single run configuration with measurements
3. Implement pairwise protocols
4. Implement Multi party protocols
5. Workload scenarios
6. Metrics computation
7. Calibration based on pairwise protocols
8. Run experiments and store results in dataset
9. Results analysis

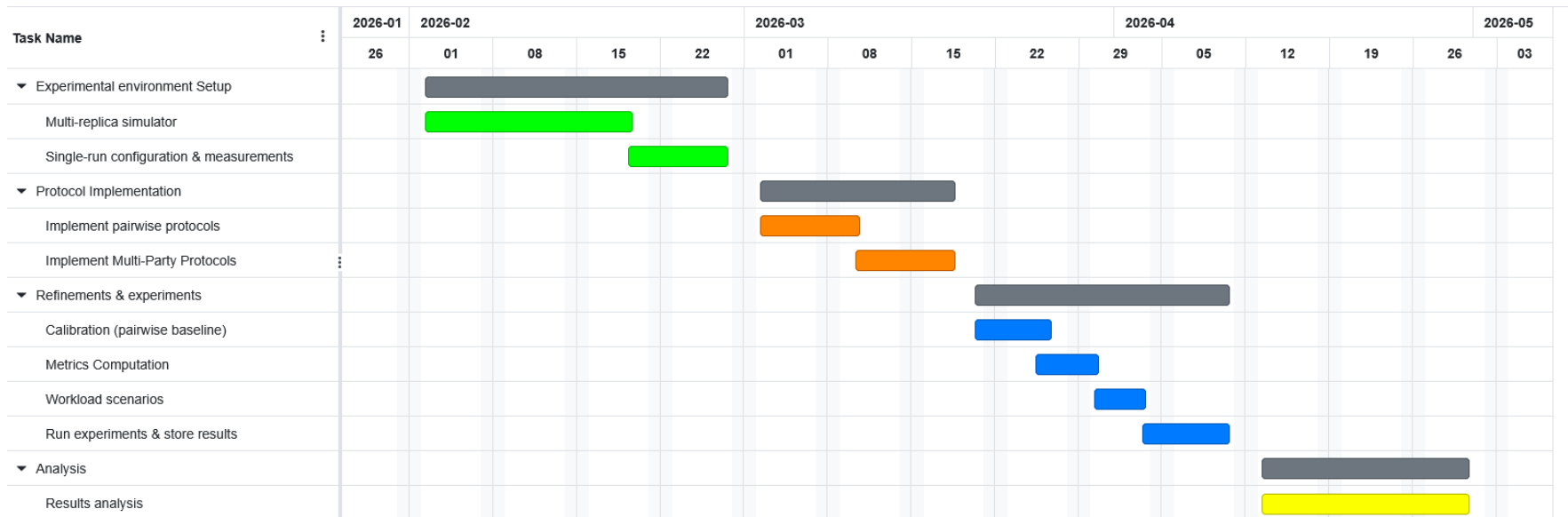


Figure 1.1: Work plan Gantt Chart

References

- [1] Pedro Silva Gomes and Carlos Baquero. *Rateless Bloom Filters: Set Reconciliation for Divergent Replicas with Variable-Sized Elements*. arXiv:2510.27614 [cs]. Oct. 2025. DOI: [10.48550/arXiv.2510.27614](https://doi.org/10.48550/arXiv.2510.27614). URL: <http://arxiv.org/abs/2510.27614> (visited on 01/16/2026).